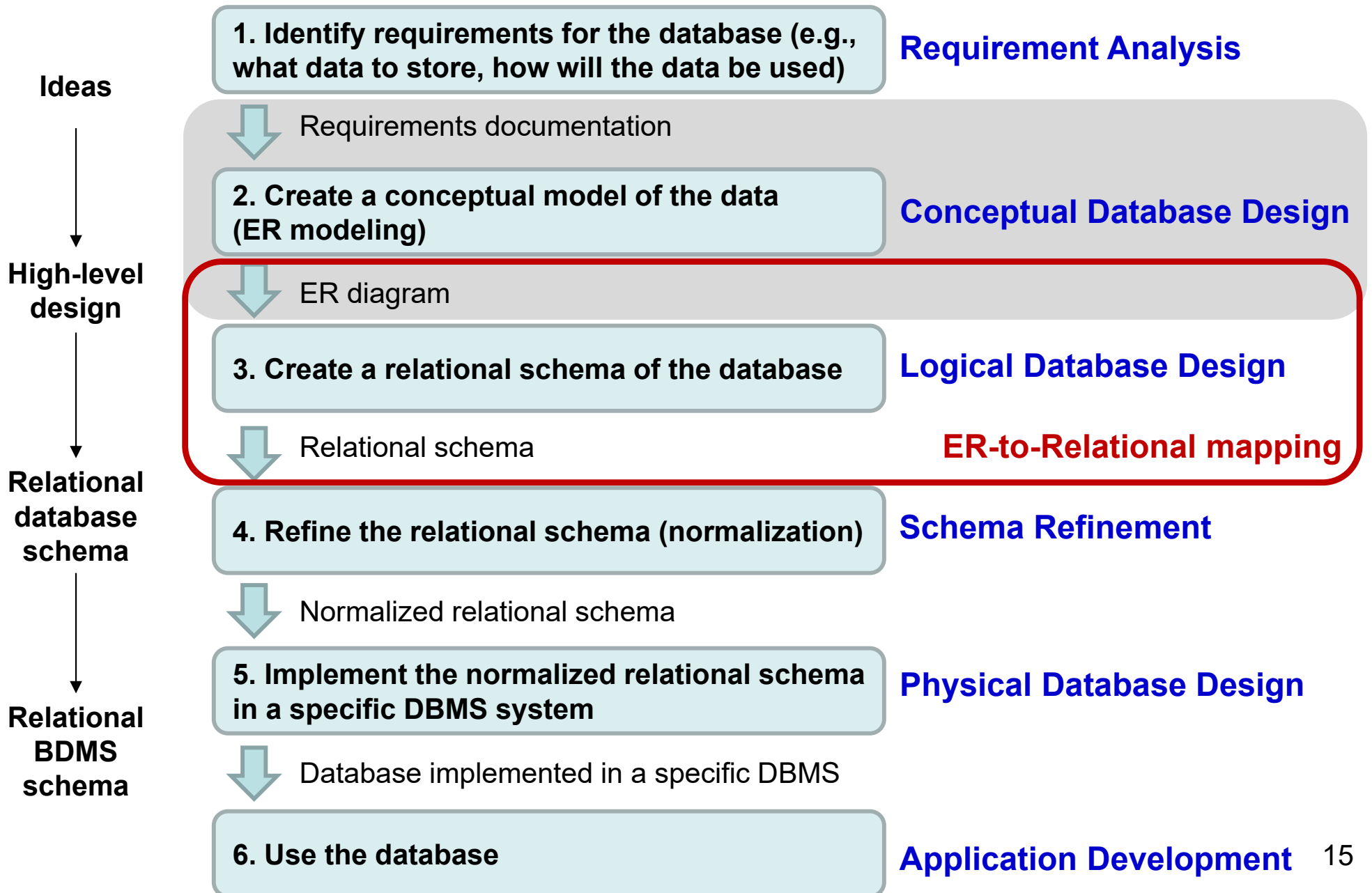
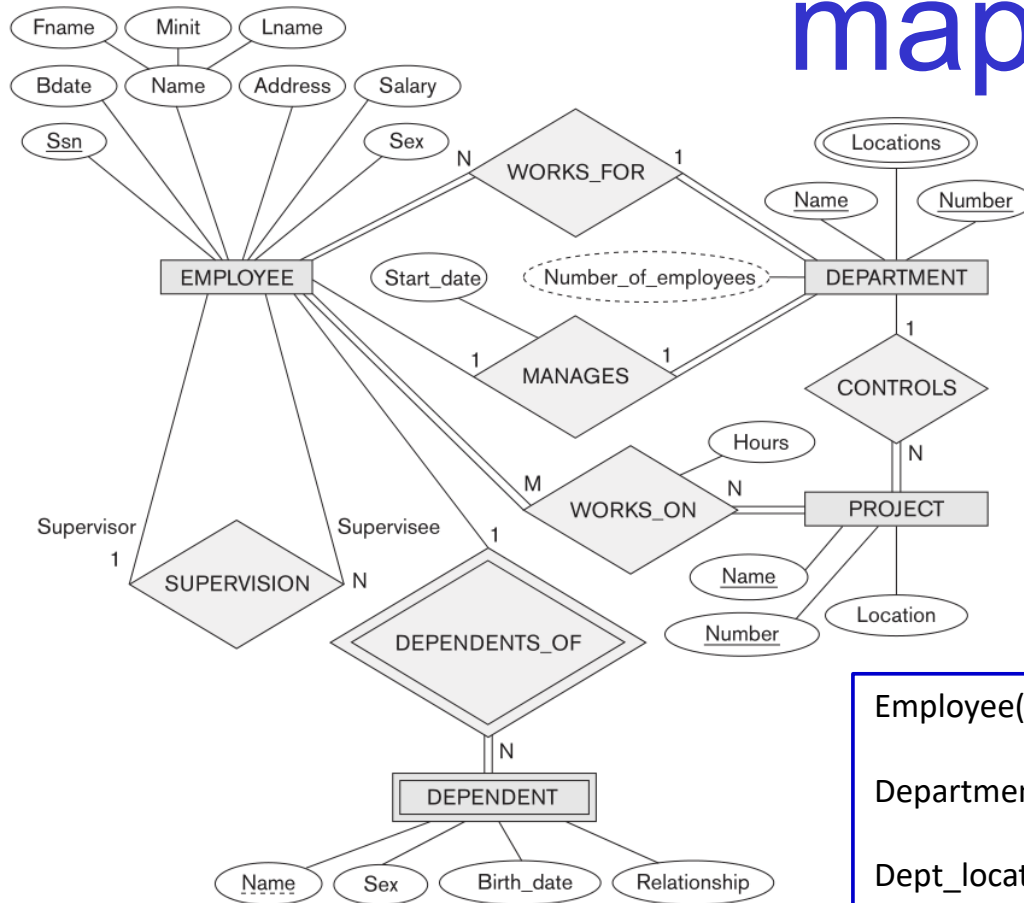


ER-to-Relational mapping

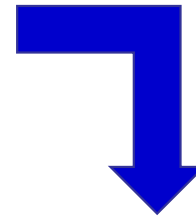
Building a database



Example ER-to-Relational mapping



Going from an ER model to a relational model (understandable by a relational DBMS)



Employee(Fname, Minit, Lname, Ssn, Bdate, Address, Sex, Salary, Super_ssn, Dno)

Department(Dname, Dnumber, Mgr_ssn, Mgr_start_date)

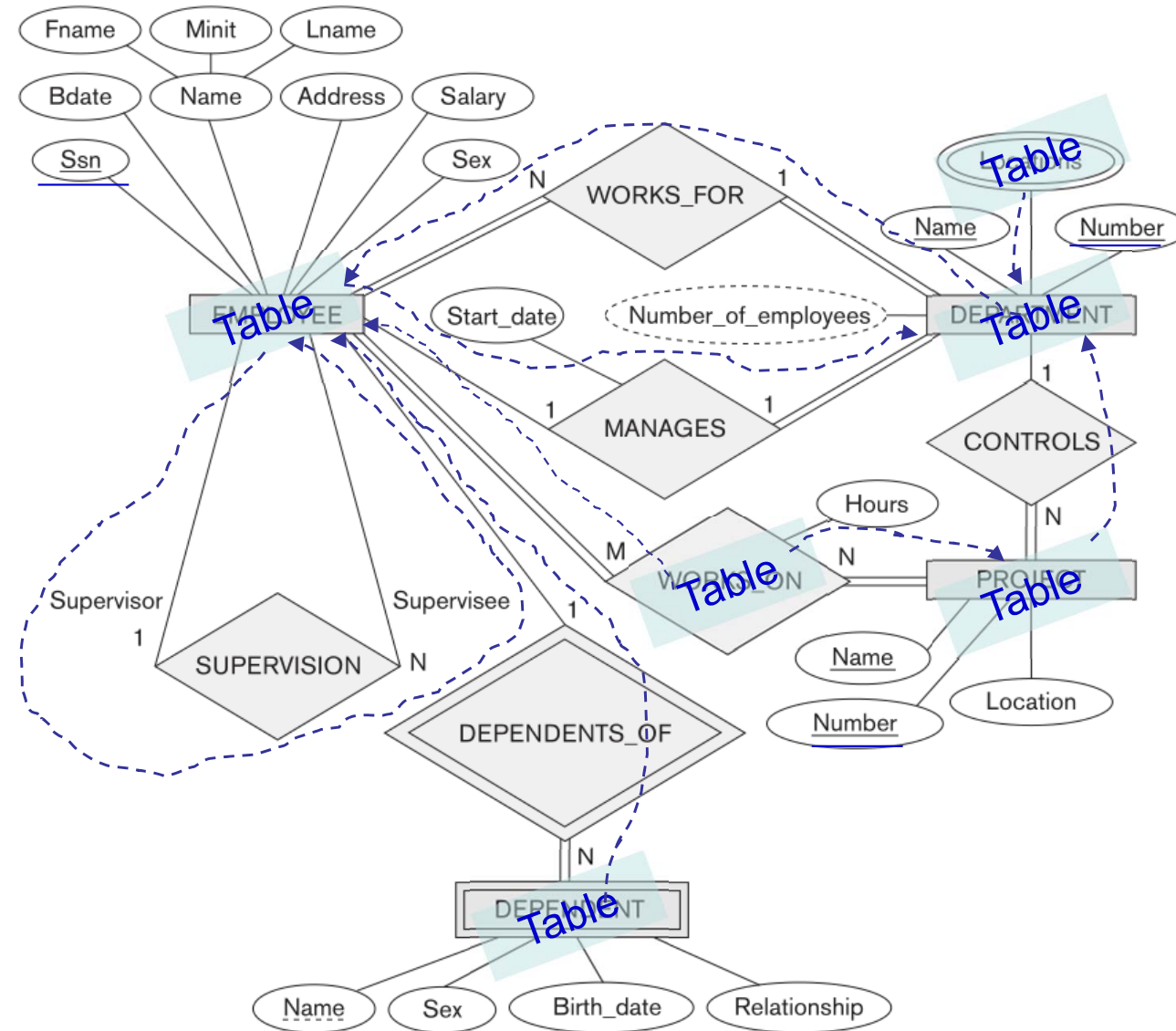
Dept_locations(Dnumber, Dlocation)

Project(Pname, Pnumber, Plocation, Dnum)

Works_on(Essn, Pno, Hours)

Dependent(Essn, Dependent_name, Sex, Bdate, Relationship)

ER-to-Relational mapping overview



Overview:

- Entities become relations (i.e., tables)
- Simple attributes and elements of composite attributes become attributes in relations (i.e., column headers in tables); key attributes become candidate keys
- Complex attributes become tables with foreign keys back to the parent entity table (its primary key)
- one-to-one or one-to-many relationships imply connections by primary key/foreign key pairs between tables
- many-to-many relationships become an “interconnect” table that simulates two equivalent one-to-many relationships
- Ternary relationships become an “interconnect” table with primary key/foreign key pairs to simulate actual relationships among attributes

ER-to-Relational mapping: A seven-step mapping procedure

Step 1: Mapping of Regular Entity Types



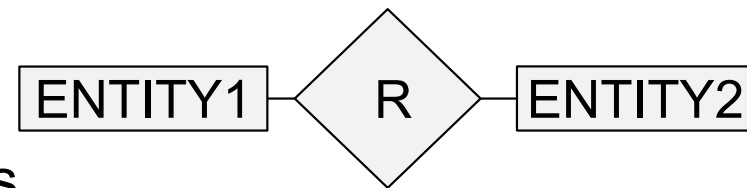
Step 2: Mapping of Weak Entity Types



Step 3: Mapping of Binary 1:1 Relation Types

Step 4: Mapping of Binary 1:N Relationship Types

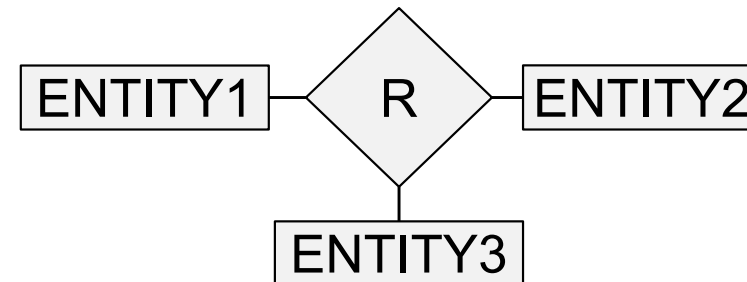
Step 5: Mapping of Binary M:N Relationship Types



Step 6: Mapping of Multivalued attributes



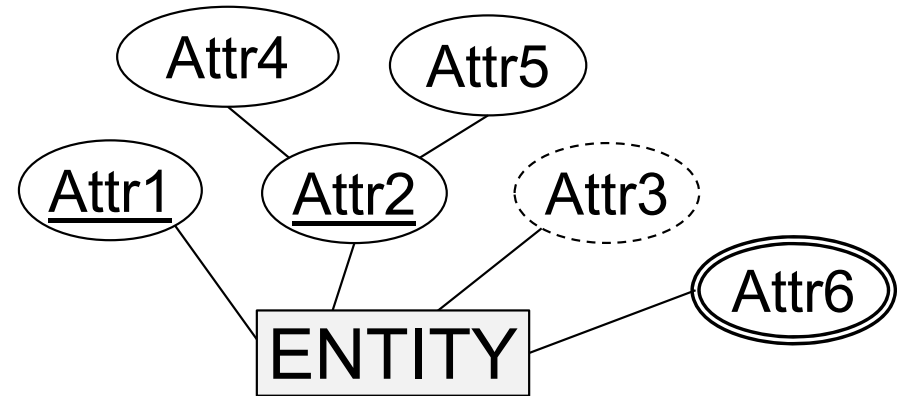
Step 7: Mapping of Ternary Relationship Types



Step 1: Mapping of Regular Entity Types

For each regular entity type:

1. Create a relation
2. For every simple attribute of the entity type, create an attribute of the relation
 - For composite attributes only their constituent atomic components are recorded
 - Derived attributes are not recorded
 - Multivalued attributes are treated later (Step 6)
3. For every key of the entity type create candidate keys in the relation, and choose a primary key



Rel(Attr1, Attr4, Attr5)

Step 1 example



Employee(Fname, Minit, Lname, Ssn, Bdate, Address, Sex, Salary)

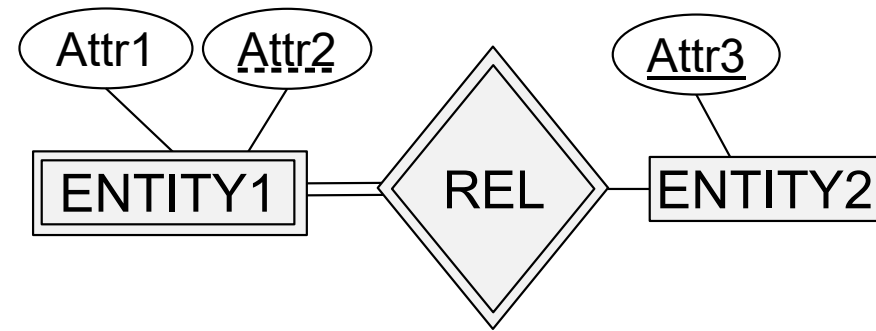
Department(Dname, Dnumber)

Project(Pname, Pnumber, Plocation)

Step 2: Mapping of Weak Entity Types

For each weak entity type:

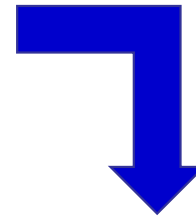
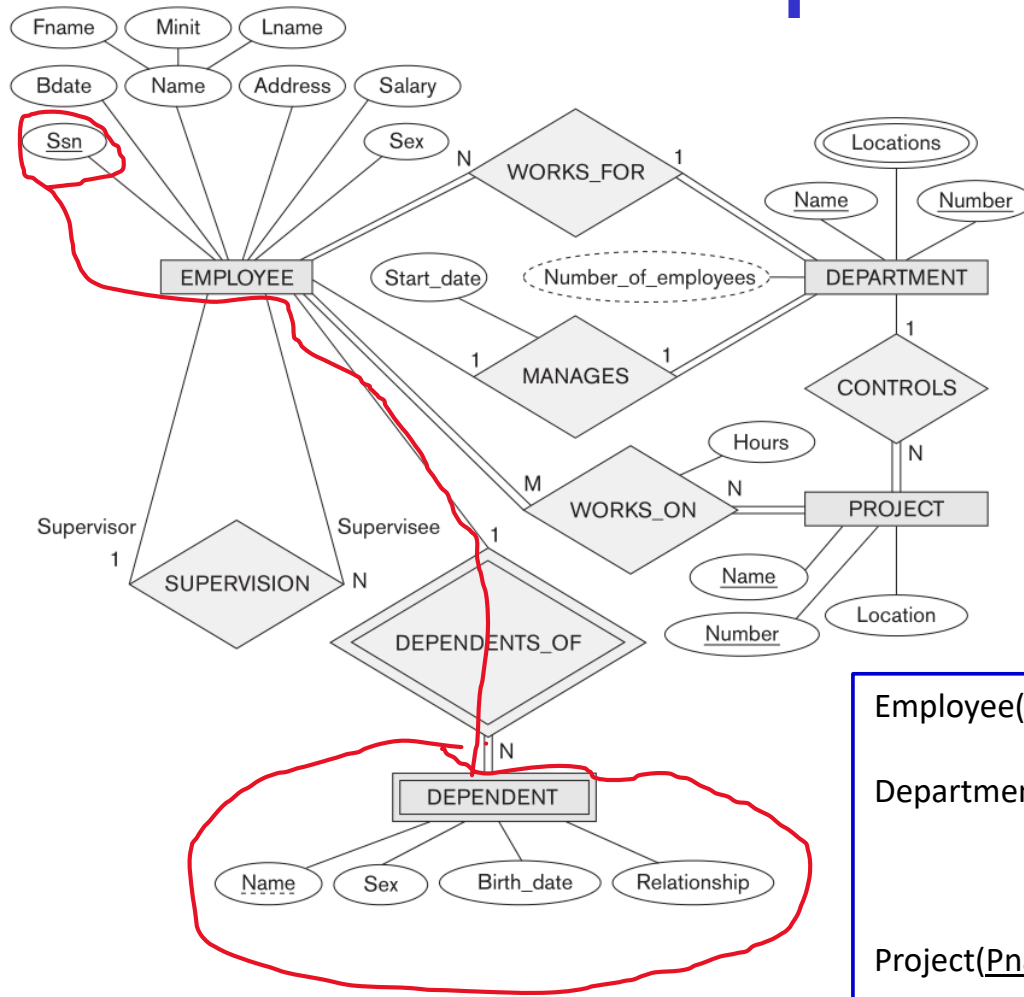
1. Create a relation
2. For every simple attribute of the entity type, create an attribute of the relation
 - For composite attributes only their constituent atomic components are recorded
3. Add the key(s) of the identifying (parent) entity type as attribute(s) in the relation
 - Include the added attribute(s) as foreign key(s) corresponding to the key attribute(s) of the parent entity type
4. The attribute(s) thus added plus the partial key(s) of the weak entity type form the primary key of the relation representing the weak entity type



RelE1(Attr1, Attr2, Attr3)

RelE2(Attr3)

Step 2 example



Employee(Fname, Minit, Lname, Ssn, Bdate, Address, Sex, Salary)

Department(Dname, Dnumber)

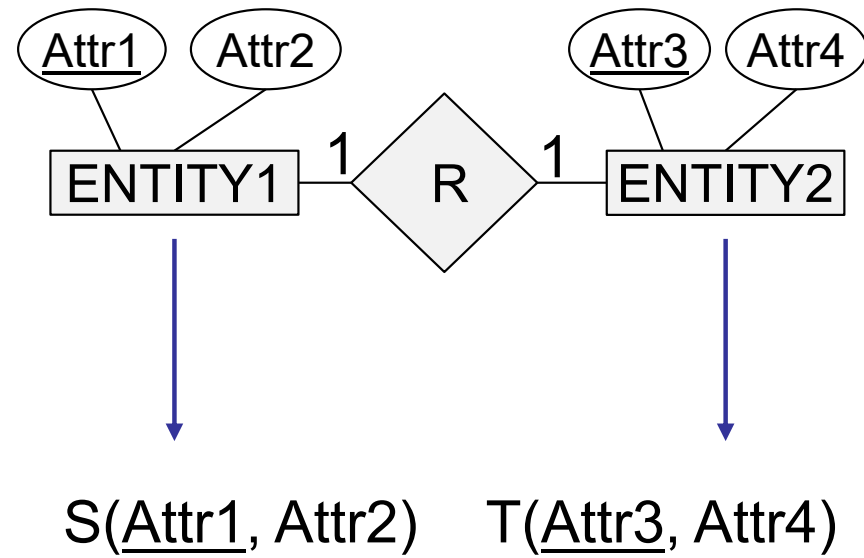
Project(Pname, Pnumber, Plocation)

Dependent(Essn, Dependent_name, Sex, Bdate, Relationship)

Step 3: Mapping of Binary 1:1 Relation Types

For each binary 1:1 relationship type R:

1. Identify the relations S and T that correspond to the entity types participating in R
2. Choose one of the three possible approaches:
 1. Option 1: Foreign key approach
 2. Option 2: Merged relationship approach
 3. Option 3: Relationship relation approach



Note: Option 1 should be followed unless special conditions exist (Options 2 and 3)

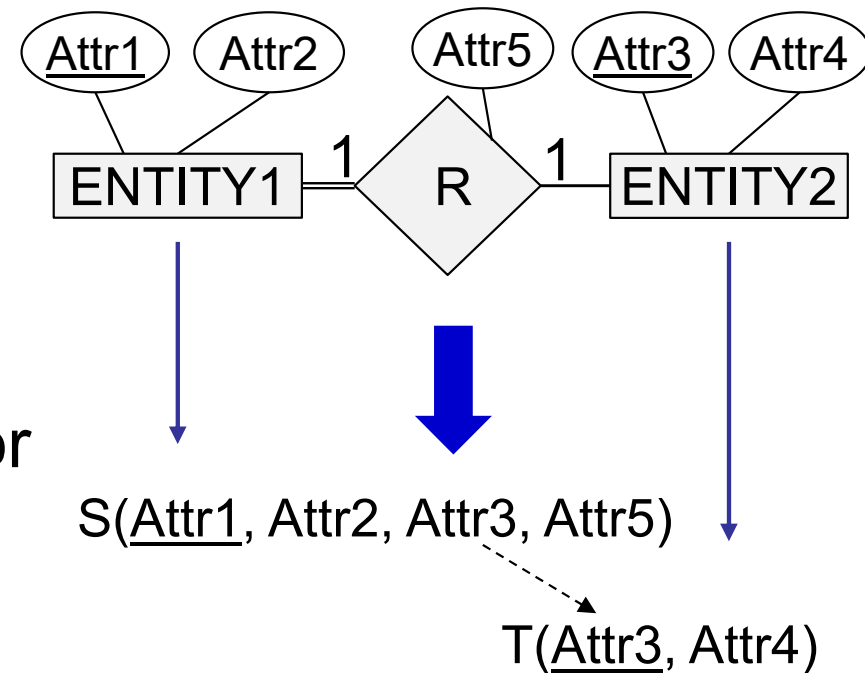
Step 3 Option 1

(Foreign key approach)

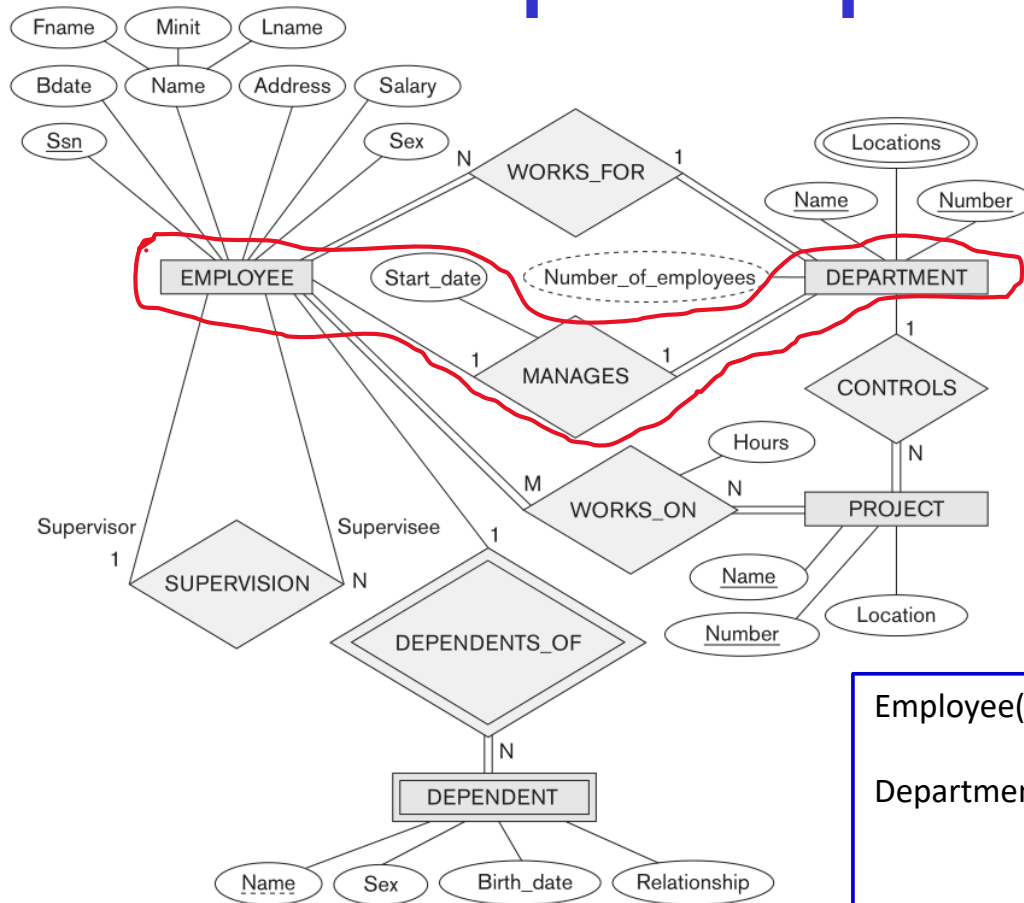
a. Choose one of the relations (e.g., S) and include as a foreign key in S the primary key of T

- If one of the entity types has total participation in R, choose that one in the role of S

b. Include all the simple attributes (or simple components of composite attributes) of the 1:1 relationship type R as attributes of S



Step 3 Option 1 example



Employee(Fname, Minit, Lname, Ssn, Bdate, Address, Sex, Salary)

Department(Dname, Dnumber, Mgr_ssn, Mgr_start_date)

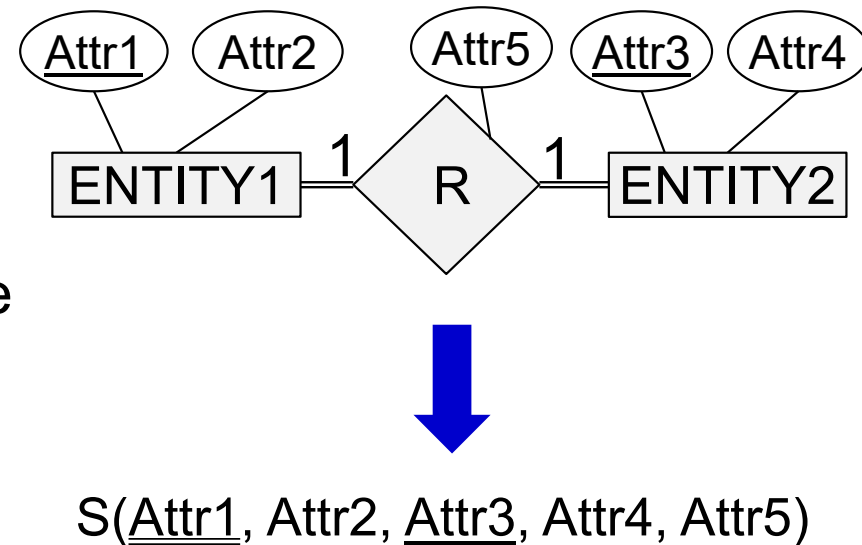
Project(Pname, Pnumber, Plocation)

Dependent(Essn, Dependent_name, Sex, Bdate, Relationship)

Step 3 Option 2

(Merged relationship approach)

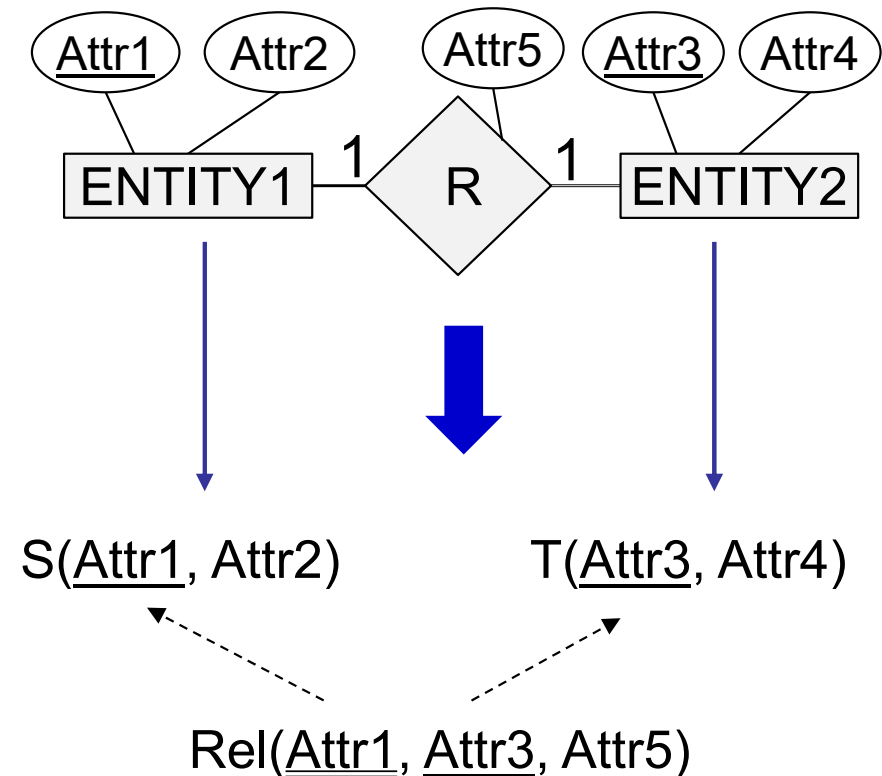
- a. When both entity types have total participation in R, merge the two entity types and the relationship into a single relation
 - The two relations corresponding to the two entity types will have the exact same number of tuples at all times, thus they can be merged into one
- b. Choose one of the keys of the two entity types as a primary key in the resulting relation



Step 3 Option 3

(Relationship relation approach)

- Set up a third relation Rel (a.k.a. the “relationship relation”) for the purpose of cross-referencing the primary keys of the two relations S and T representing the entity types
- The relation Rel include in the primary key of relation Rel the attributes of S and T as foreign keys to S and T; and add to the attributes of Rel any other attributes of the relationship type R
- The primary key of Rel will be one of the two foreign keys, and the other foreign key will be a candidate key of Rel



Step 4: Mapping of Binary 1:N Relationship Types

Two possible approaches:

1. Option 1: Foreign key approach
2. Option 2: Relationship relation approach

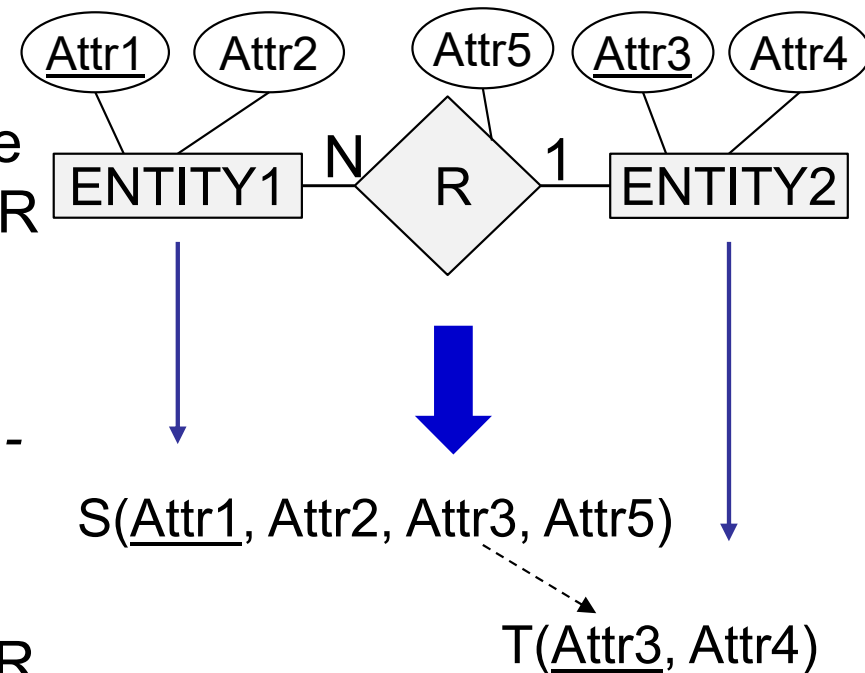
Note: Option 1 is generally preferred as it reduces the number of resulting relations

Step 4 Option 1

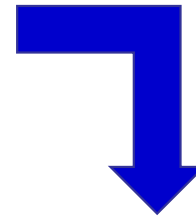
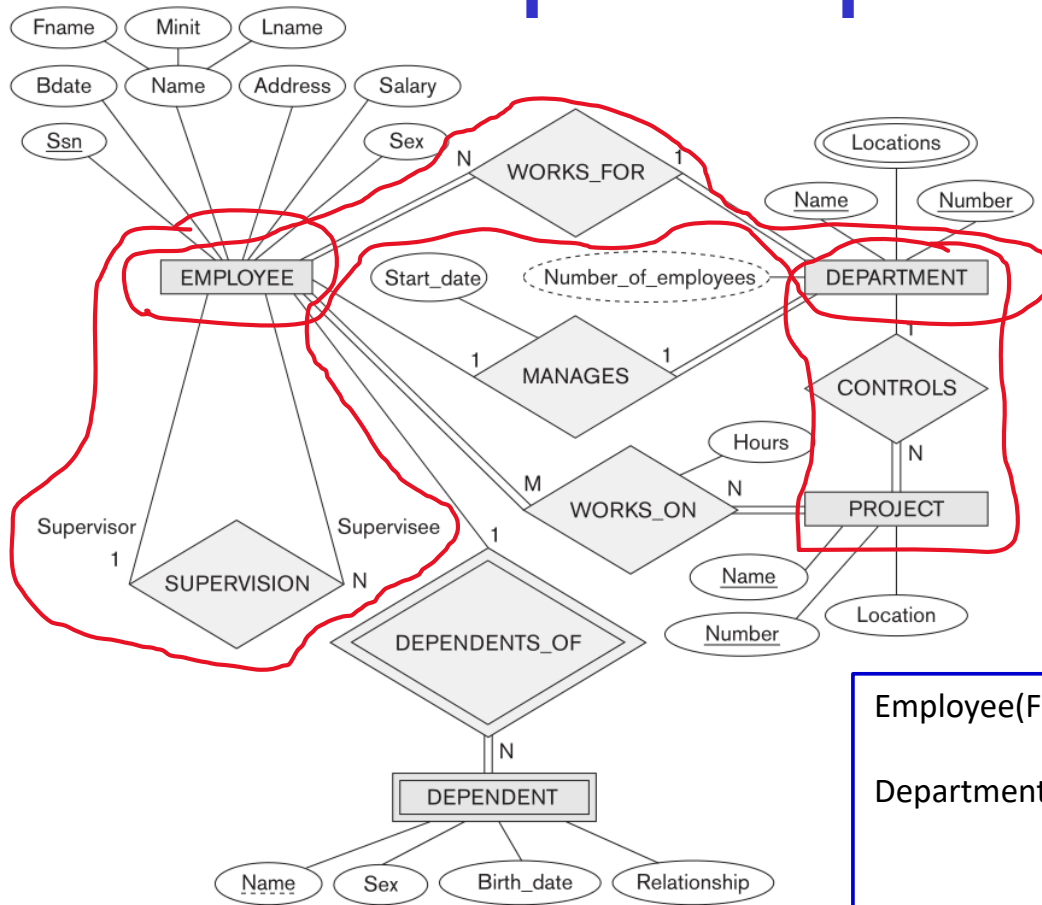
(Foreign key approach)

For each regular binary 1:N (or N:1) relationship type R:

- Identify the relation S that represents the participating entity type at the *N-side* of R
- Include as foreign key in S the primary key of the relation T that represents the other entity type participating in R (the *1-side* of R)
- Include any simple attributes (or simple components of composite attributes) of R as attributes of S



Step 4 Option 1 example



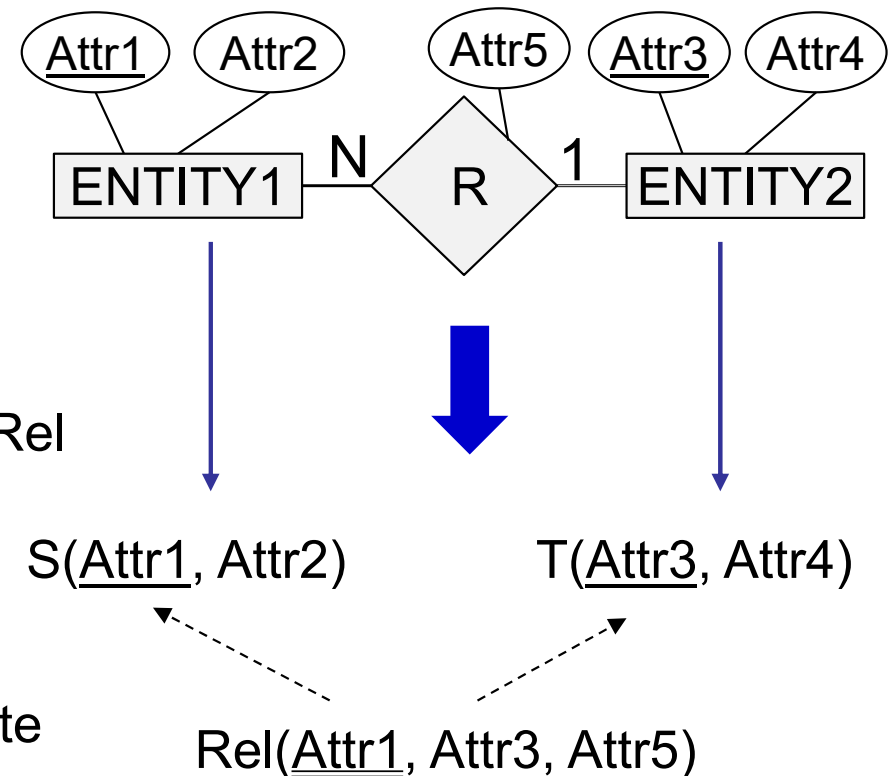
Employee(Fname, Minit, Lname, Ssn, Bdate, Address, Sex, Salary, Super_ssn, Dno)
 Department(Dname, Dnumber, Mgr_ssn, Mgr_start_date)
 Project(Pname, Pnumber, Plocation, Dnum)
 Dependent(Essn, Dependent_name, Sex, Bdate, Relationship)

Step 4 Option 2

(Relationship relation approach)

For each regular binary 1:N (or N:1) relationship type R:

- Identify the relation S that represents the participating entity type at the *N-side* of the relationship type
- Create a separate relation Rel whose attributes are the primary keys of S and T, which will also be foreign keys to S and T; include any attributes of R as attributes of Rel
- The primary key of Rel is the same as the primary key of S

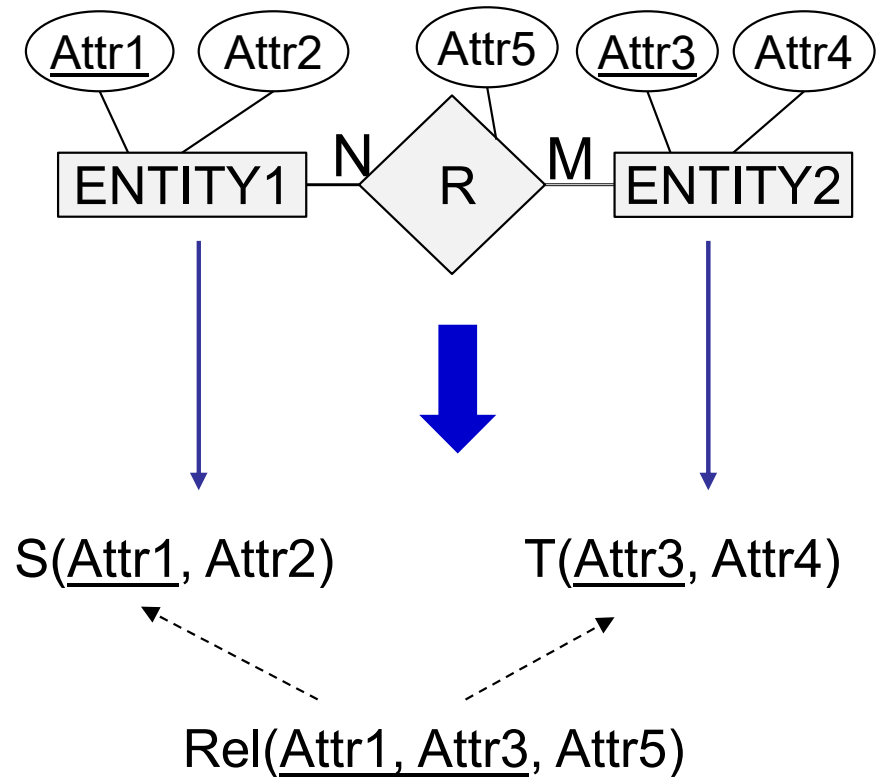


Note: Use this option if few tuples in S participate in the relationship to avoid excessive NULL values in the foreign key

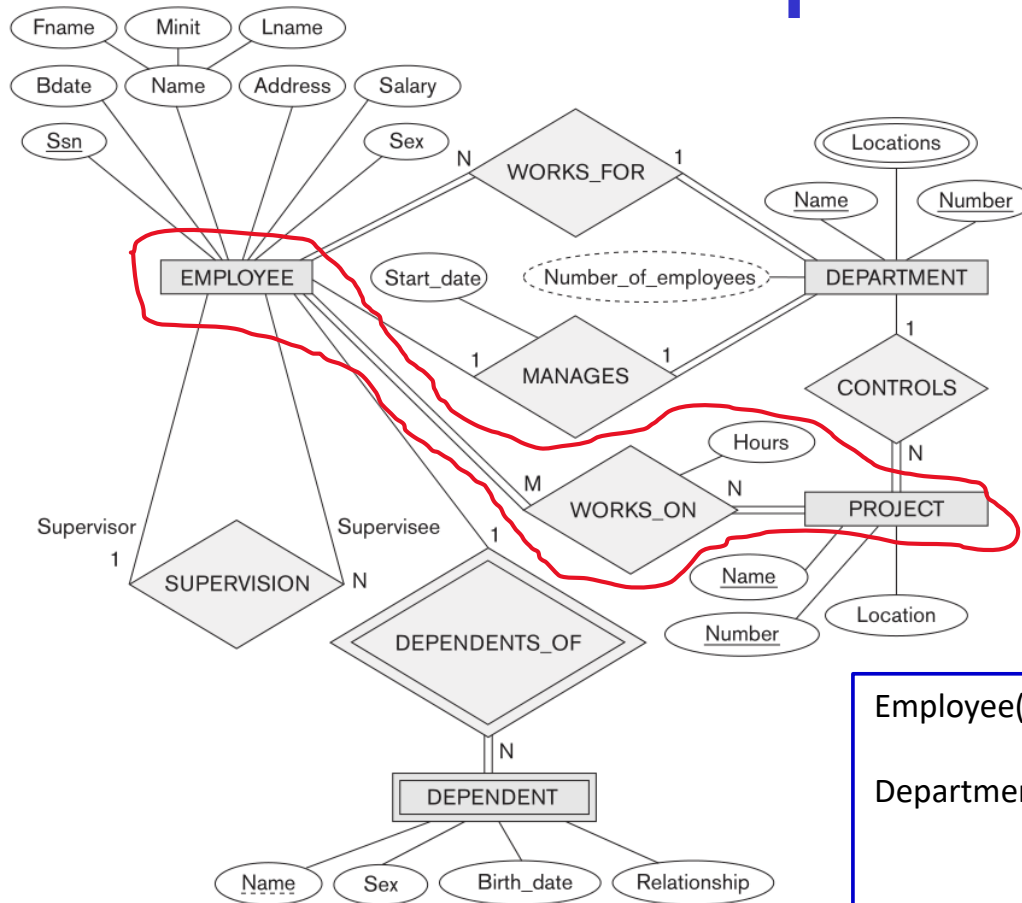
Step 5: Mapping of Binary M:N Relationship Types

For each binary M:N relationship type R:

1. Create a new relation Rel to represent R
2. Include as foreign key attributes in Rel the primary keys of the relations that represent the participating entity types (S and T); their combination will form the primary key of Rel
3. Also include any simple (or simple components of composite) attributes of R as attributes of Rel



Step 5 example



Employee(Fname, Minit, Lname, Ssn, Bdate, Address, Sex, Salary, Super_ssn, Dno)

Department(Dname, Dnumber, Mgr_ssn, Mgr_start_date)

Project(Pname, Pnumber, Plocation, Dnum)

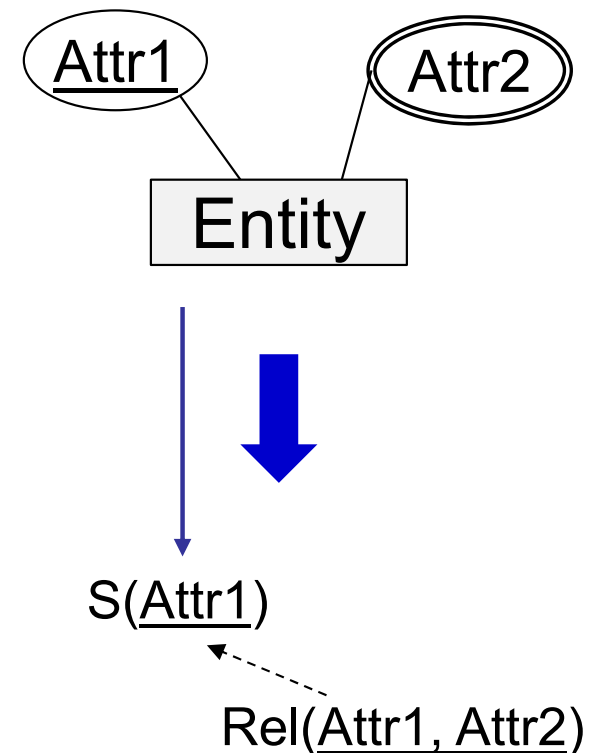
Works_on(Essn, Pno, Hours)

Dependent(Essn, Dependent_name, Sex, Bdate, Relationship)

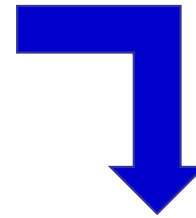
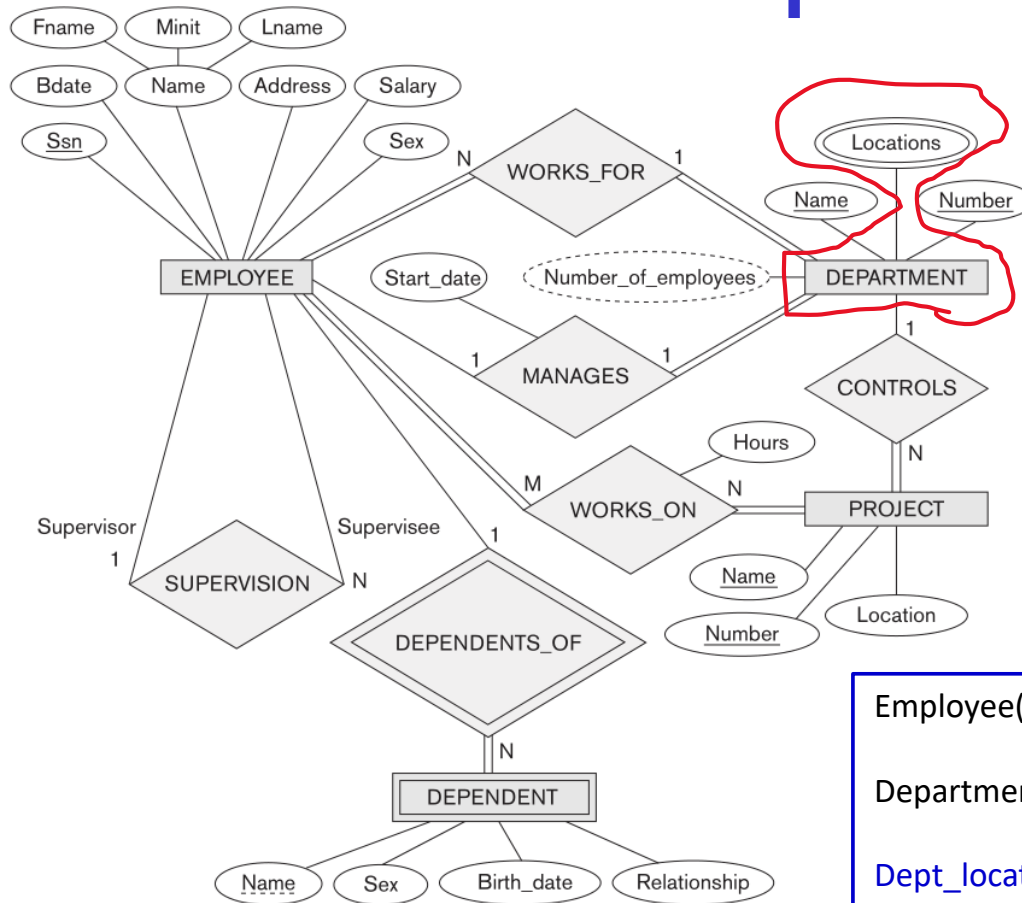
Step 6: Mapping of Multivalued attributes

For each multivalued attribute A:

1. Create a new relation Rel
2. Include an attribute corresponding to A, plus the primary key attribute K—as a foreign key in Rel—of the relation that represents the entity type (or relationship type) that has A as a multivalued attribute
3. The primary key of Rel is the combination of A and K
4. If the multivalued attribute is composite, include its simple components



Step 6 example



Employee(Fname, Minit, Lname, Ssn, Bdate, Address, Sex, Salary, Super_ssn, Dno)

Department(Dname, Dnumber, Mgr_ssn, Mgr_start_date)

Dept_locations(Dnumber, Dlocation)

Project(Pname, Pnumber, Plocation, Dnum)

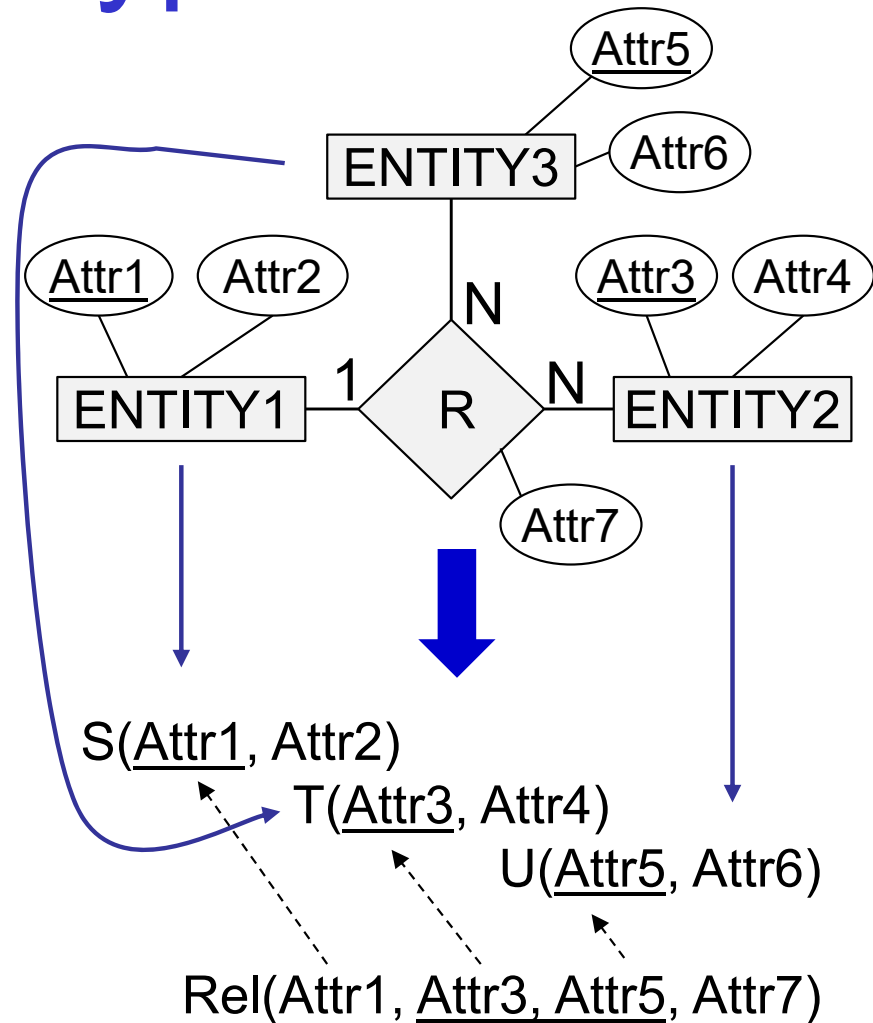
Works_on(Essn, Pno, Hours)

Dependent(Essn, Dependent_name, Sex, Bdate, Relationship)

Step 7: Mapping of Ternary Relationship Types

For each ternary relationship type R:

- Create a new relationship relation Rel to represent R
- Include as foreign key attributes in Rel the primary keys of the relations that represent the participating entity types (S, T, U); also include any simple (or simple components of) attributes of R as attributes of Rel
- Candidate keys in Rel are established as follows – if S:T:U is:
 - 1:1:1: three candidate keys formed by the combination of primary keys of (S,T), (S,U), and (T,U) respectively
 - 1:1:N: two candidate keys formed by the combination of primary keys of (S,U), and (T,U) respectively
 - 1:N:N: one candidate key formed by the combination of primary keys of (T,U)
 - N:N:N: one candidate key formed by the combination of primary keys of (S, T,U)



Correspondence between ER and relational models

ER Model

Entity type

1:1 or 1:N relationship type

M:N relationship type

n-ary relationship type

Simple attribute

Composite attribute

Multivalued attribute

Value set

Key attribute

Relational Model

“Entity” relation

Foreign key (or “relationship” relation)

“Relationship” relation and two foreign keys

“Relationship” relation and *n* foreign keys

Attribute

Set of simple component attributes

Relation and foreign key

Domain

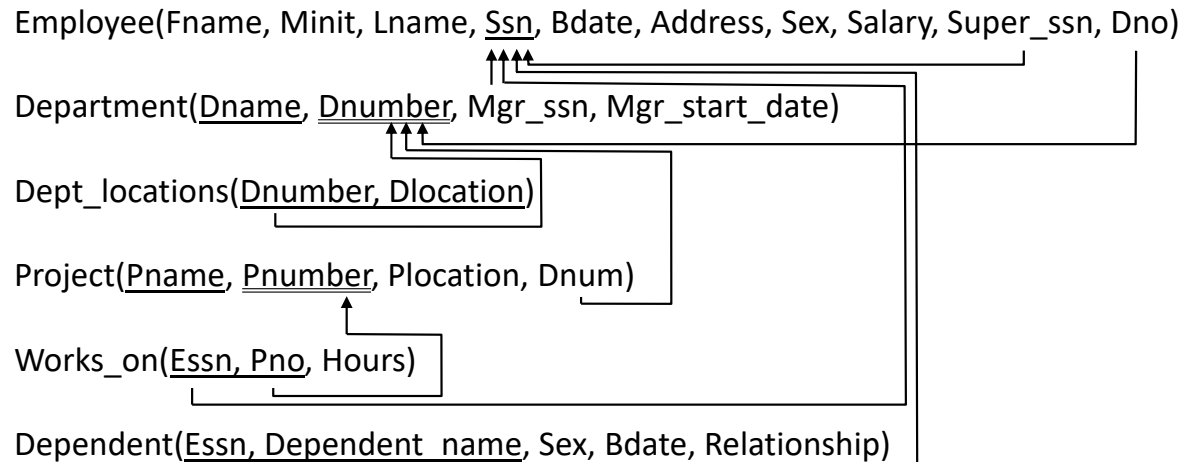
Primary (or secondary) key

Key difference between ER and relational models

- **Relationship types are not represented explicitly in the relational model**, but implicitly by having two attributes A and B, one a primary key and the other a foreign key (over the same domain) included in two relations S and T, respectively
- Two tuples in S and T are related when they have the same value for A and B
- By using the EQUIJOIN operation over S.A and T.B, we can combine all pairs of related tuples from S and T and materialize the relationship
 - M:N implies two joins
 - 1:N implies one or two joins
 - 1:1 implies zero, one or two joins
 - N-ary implies N joins

When multiple relationships need to be traversed, numerous join operations must be specified. The user must always be aware of the foreign key attributes in order to use them correctly in combining related tuples from two or more relations.

Example



Create a relation that includes the employee name, project name, and hours that the employee works on each project

$\pi_{\text{Fname, Minit, Lname, Pname, Hours}}((\text{Employee} \bowtie_{\text{Ssn=Essn}} \text{Works_on}) \bowtie_{\text{Pno=Pnumber}} \text{Project}))$

- Does it make sense to join the Project and Dept_locations relations on the condition Dlocation=Plocation ?

Summary ER modeling

- Conceptual database design is essential in building DBs
- Conceptual design follows requirements analysis
- ER is a popular choice for the conceptual design phase (some systems use the ER model for relational schema design)
- Basic ER concepts: entities, relationships, attributes, constraints
- ER models are usually easier to understand than relational models
- ER models express relationships explicitly
- Procedure to convert ER diagrams to relational schemas